

Postdoctoral researcher in Machine Learning and biomedical informatics, with research spanning clustering, high-dimensional data analysis, ensemble learning, and applied AI for biomedical and real-world data.

Author of journal, conference, and software publications, with experience in research projects, university teaching, and scientific reviewing.

EXPERIENCE

University of Thessaly

Lamia, GRC

Post Doctoral Researcher in Computer Science and Biomedical Informatics

February 2024–Current

- Advisor: Tasoulis Sotiris
- Title: “Machine Learning Algorithms in Big Data”

Institute for Bio-economy and Agri-technology iBO/CERTH

Lamia, GRC

Role: Researcher

December 2023–Current

- “Personalized Rehabilitation Via Novel AI Patient Stratification Strategies (PREPARE)”
- Defined requirements to guide the technological development of PREPARE tools.
- Developed technological tools for robust, transparent, and accurate computational models supporting patient prediction and stratification strategies to improve clinical outcomes.

University of Thessaly

Lamia, GRC

Role: Researcher

June 2024–December 2025

- “Bridging big omic, genetic and medical data for Precision Medicine implementation in Greece (project code TAEDR-0539180)”
- Contributed to the development of methods for integrating large-scale omics data.
- Developed machine-learning methods using statistical and graph-based systems-biology approaches.
- Contributed to linking machine-learning methods and large-scale omics data with an intelligent electronic health-record system and to the development of an intelligent patient-registry mechanism.

Institute for Bio-economy and Agri-technology iBO/CERTH

Lamia, GRC

Role: Researcher

August 2021–November 2023

- “uPrevent”
- Developed software and evaluated the validity of the system.

Signal Ocean SMPC

Lamia, GRC

Role: Researcher

March 2021–December 2021, February 2023–October 2023

- “Design and implementation of machine learning algorithms for the prediction and analysis of the course of transport vessels”
- Designed, evaluated, and delivered fuzzy string-matching methods for entity-name resolution, combining edit-distance, dictionary-based, phonetic, and tree-based approaches; the developed solution was incorporated into the company’s production workflow.

University of Thessaly

Lamia, GRC

Role: Researcher

March 2022–September 2023

- “ParICT_CENG: Improving ICT research infrastructures in Central Greece for processing large volumes of data from sensor streams, multimedia and complex mathematical simulation models”

- Participated in WP1, contributing to the development, operation, and maintenance of a high-performance parallel computing cluster.

Institute for Bio-economy and Agri-technology iBO/CERTH

Lamia, GRC

Role: Researcher

March 2020–April 2021

- “Advanced personalised, multi-scale computer models preventing OsteoArthritis,”
- Developed and validated computational, machine-learning, and deep-learning models using heterogeneous data.

EDUCATION

University of Thessaly

Lamia, GRC

Doctor of Philosophy (Ph.D.) in Computer Science and Biomedical Informatics

2019–2023

- Thesis: “Design and Implementation of Machine Learning Algorithms in Big Biomedical Data”

University of Thessaly

Lamia, GRC

Certificate of Pedagogical & Teaching Competence (Παιδαγωγικής & Διδακτικής Επάρκειας)

2022–2023

University of Patras

Patras, GRC

Diploma in Computer Engineering and Informatics.

2019

- Thesis: “Knowledge mining and visualization of molecular biology networks”
- Bachelor with integrated master at the Polytechnic School of University of Patras.

PUBLICATIONS

- [1] **P. Anagnostou**, S. K. Tasoulis, A. G. Vrahatis, S. V. Georgakopoulos, and V. P. Plagianakos, “Boosting neural network performance for high dimensional data through random projections”, *Pattern Recognition Letters*, vol. 199, pp. 149–155, 2026, ISSN: 0167-8655.
- [2] P. Barmpas, **P. Anagnostou**, S. K. Tasoulis, V. P. Plagianakos, and S. V. Georgakopoulos, “Efficient hybrid hierarchical clustering with incremental silhouette score for large, noisy datasets”, *International Journal of Neural Systems*, 2025.
- [3] A. Biloni, **P. Anagnostou**, S. K. Tasoulis, S. V. Georgakopoulos, and V. P. Plagianakos, “Enhancing machine learning models for medical coding: Diagnostic codes mapping and synthetic clinical notes”, in *2025 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, IEEE, 2025, pp. 1–8.
- [4] P. C. Theocharopoulos, **P. Anagnostou**, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Large language models for efficient topic modeling”, *Neural Computing and Applications*, pp. 1–19, 2025.
- [5] K. Tsiaktani, **P. Anagnostou**, S. K. Tasoulis, S. V. Georgakopoulos, J. A. Gittings, D. E. Raitsos, A. Sergounioti, and V. P. Plagianakos, “Short-term impacts of weather conditions on biomarkers among urban adults in greece”, in *2025 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, 2025, pp. 1–8.
- [6] P. Barmpas, **P. Anagnostou**, S. K. Tasoulis, S. V. Georgakopoulos, and V. P. Plagianakos, “HCER: Hierarchical clustering-ensemble regressor”, in *Engineering Applications of Neural Networks*, L. Iliadis, I. Maglogiannis, A. Papaleonidas, E. Pimenidis, and C. Jayne, Eds., Cham: Springer Nature Switzerland, 2024, pp. 369–378, ISBN: 978-3-031-62495-7.
- [7] **P. Anagnostou**, “Design and implementation of machine learning algorithms in big biomedical data”, English, ND: 56040, PhD thesis, University of Thessaly, 2023.

- [8] **P. Anagnostou**, P. Barmpas, S. K. Tasoulis, S. V. Georgakopoulos, and V. P. Plagianakos, “Neural networks voting for projection based ensemble classifiers”, in *2023 IEEE International Conference on Big Data (BigData)*, IEEE, 2023, pp. 4567–4574.
- [9] **P. Anagnostou**, N. G. Pavlidis, and S. K. Tasoulis, “Ensemble clustering for boundary detection in high-dimensional data”, in *Proceedings of the 9th Annual Conference on Machine Learning, Optimization and Data Science (LOD)*, Lake District, UK: Springer, 2023.
- [10] **P. Anagnostou**, S. Tasoulis, V. P. Plagianakos, and D. Tasoulis, “HiPart: Hierarchical divisive clustering toolbox”, *Journal of Open Source Software*, vol. 8, no. 84, p. 5024, 2023.
- [11] P. C. Theocharopoulos, **P. Anagnostou**, A. Tsoukala, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Detection of fake generated scientific abstracts”, in *2023 IEEE Ninth International Conference on Big Data Computing Service and Applications (BigDataService)*, 2023, pp. 33–39.
- [12] P. Barmpas, S. Tasoulis, A. G. Vrahatis, S. V. Georgakopoulos, **P. Anagnostou**, M. Prina, J. L. Ayuso-Mateos, J. Bickenbach, I. Bayes, M. Bobak, *et al.*, “A divisive hierarchical clustering methodology for enhancing the ensemble prediction power in large scale population studies: The athlos project”, *Health Information Science and Systems*, vol. 10, no. 1, p. 6, 2022.
- [13] **P. Anagnostou**, S. Tasoulis, A. G. Vrahatis, S. Georgakopoulos, M. Prina, J. L. Ayuso-Mateos, J. Bickenbach, I. Bayes-Marin, F. F. Caballero, L. Egea-Cortés, *et al.*, “Enhancing the human health status prediction: The athlos project”, *Applied Artificial Intelligence*, vol. 35, no. 11, pp. 834–856, 2021.
- [14] **P. Anagnostou**, P. Barbas, A. G. Vrahatis, and S. K. Tasoulis, “Approximate knn classification for biomedical data”, in *2020 IEEE International Conference on Big Data (Big Data)*, IEEE, 2020, pp. 3602–3607.
- [15] **P. Anagnostou**, “Knowledge mining and visualization of molecular biology networks”, Graduate Thesis, University of Patras, School of Engineering, Oct. 2019.

INVITED TALKS

- N. G. Pavlidis, **P. Anagnostou**, and S. Tasoulis, “Dimensionality reduction and clustering”, in Proceedings 35th Panhellenic & 1st International Statistics Conference, Athens, Greece, 2023. Co-presenters: Pavlidis, Nicos G., and Tasoulis, Sotiris.

ACADEMIC SERVICE

- **Journal reviewer for:** Pattern Recognition Letters, Future Generation Computer Systems, Journal of Open Source Software, Neural Computing and Applications
- **Conference reviewer for:** IEEE Congress on Evolutionary Computation (IEEE CEC), IEEE Symposium Series on Computational Intelligence (IEEE SSCI)
- **Program Committee Member:** IEEE CIBCB (2025); IEEE BigData (2021, 2023)

TEACHING

Instructor appointments at the Department of Computer Science and Biomedical Informatics, University of Thessaly were part of the program “Acquisition of Academic Teaching Experience for New PhD Holders,” co-funded by the European Union and national resources for the periods Spring 2024–Spring 2025 and Spring 2026.

- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2026
Linked Computing and Data Systems
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2026

Ubiquitous Computing Applications Programming

- **Instructor** at Department of Mathematics, University of Thessaly Fall 2025
Programming Language I
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2025
Linked Computing and Data Systems
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2025
Ubiquitous Computing Applications Programming
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Fall 2024
Microprocessors
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Fall 2024
Internet Application Technologies
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2024
Embedded Computer Systems in Biomedical Engineering
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2024
Ubiquitous Computing Applications Programming
- **Teaching Assistant** at University of Thessaly's School of Science Fall 2019–2024
Educational Software Principles
- **Teaching Assistant** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2021–2023
Deep Learning
- **Teaching Assistant** at University of Thessaly's School of Science Spring 2020–2022
Computer Programming Subjects
- **Laboratory Assistant** at Department of Computer Engineering and Informatics, University of Patras Spring 2017
Digital Electronics Laboratory

TECHNICAL SKILLS

- **Programming Languages:** Python, R, Bash, C/C++, Java
- **Research/Data Tools:** LaTeX, MATLAB, MySQL
- **Infrastructure:** Git, Docker, Linux
- **Additional:** HTML/CSS/JavaScript, XML, Maven